Drew University - CSCI 260 - Fall 2024
HW1 - Data Representation

HW Info.: Due on 09/09/2024 (M) by 11:59pm, 42 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

1. (16 points) Perform the following additions with 8-bit binary numbers. Then, compute the decimal values of the sums using the same corresponding representations (i.e., Signed Magnitude for part (a), and etc.). **Show all your work.**
   (a) Add the Signed Magnitude representation of 27 to the Signed Magnitude representation of -27.
   (b) Add the 1s' Complement representation of 34 to the 1s' Complement representation of -34.
   (c) Add the 2's Complement representation of 59 to the 2's Complement representation of 59.
   (d) Add Unsigned binary representation of 153 to the Unsigned binary representation of 102.

2. (10 points) Compute the following logical expressions. **Show all your work.**
   (a) NOT(1011) OR NOT(1100)
   (b) NOT(1000 AND (1100 OR 0101))
   (c) NOT(NOT(1101))
   (d) (0110 OR 0000) AND 1111
   (e) NOT(xDEFA) AND NOT(xFFFF) in hexadecimal

3. (16 points) Consider four hexadecimal numbers: 0x434F, 0x4D50, 0x5554, and 0x4552. What values do they represent if treated as each of the four data types shown? For the first 3 rows, you should provide decimal values; and for the last row, you should provide text values. **Show all your work.**

	0x434F	0x4D50	0x5554	0x4552
Unsigned Binary				
1's Complement				
2's Complement				
ASCII String				